Research & Decision Intelligence Agent: Technical Documentation

Project 3, Agentic AI Final Team Project · OpenAI Agents SDK · Python 3.11+ · Gradio UI Team: **Karim Ashraf** · **Ahmed**

> **Core principle:** *the LLM decides what to do; Python decides what is allowed.* Decisions about safety, routing, publishing and stopping are made in plain Python, which can be tested without an API key.

---

1. What the system does

A user asks a business or technical **decision question**, for example: *"Should Delta University replace its legacy Campus Classic LMS with Nilo Learn for all 14 faculties starting fall 2027?"*

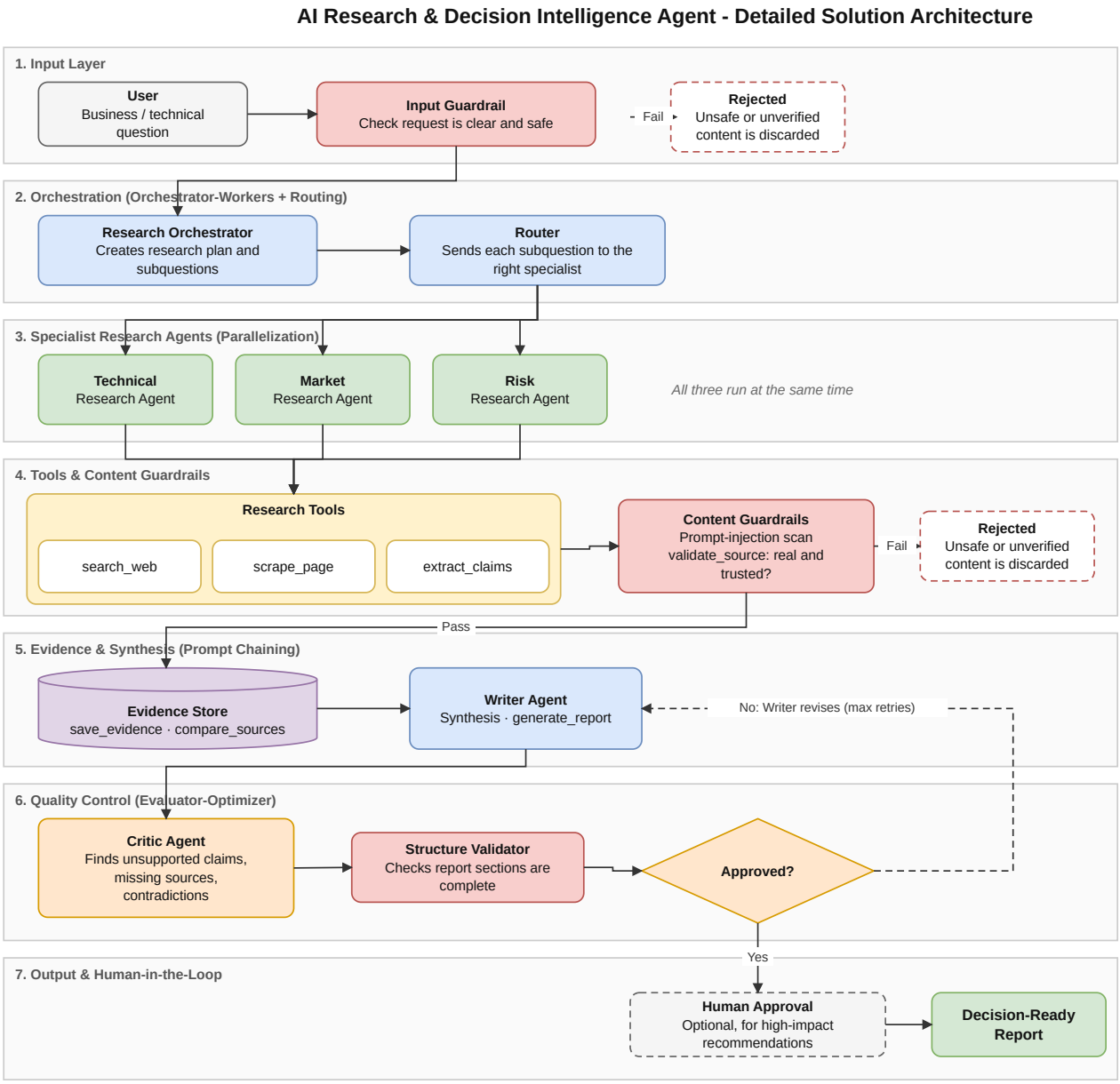
The system then:

1. **checks** the question (prompt injection, secrets, size, scope),
2. **plans** the research as sub-questions,
3. **routes** each sub-question to a specialist and adds a risk review if the plan has none,
4. **researches in parallel** (search → scrape → extract verbatim claims → save evidence),
5. **reconciles** gaps and conflicting sources with an orchestrator that delegates to workers,
6. **writes** a cited report and **critiques and revises** it,
7. **stops for a human** before publishing any high-impact or sensitive recommendation.

Every claim in the final report points to an evidence id (`E1`, `E2` ...). Each id holds a quote copied **verbatim** from a page that was actually fetched during the run.

---

2. Architecture flowchart



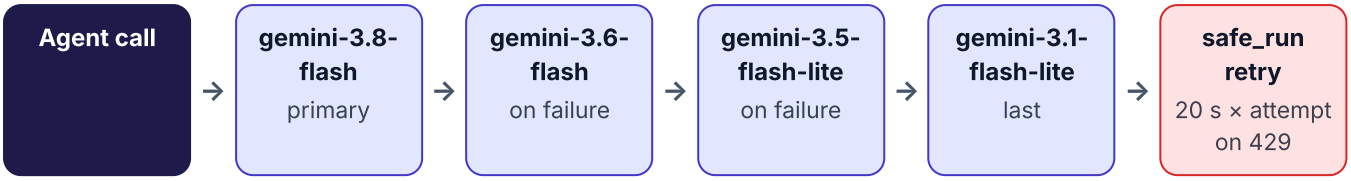
3. Inside one specialist (tool chain)

search_web
query, max_results
→
scrape_page
URL policy · retry · sanitise
→
extract_claims
verbatim sentences (Python)
→
save_evidence
rejects non-verbatim → E1, E2 …
→
SpecialistFindings
Pydantic output

A dead link is logged as `tool_failure` and the specialist moves on to the next source.

The LLM never writes an evidence id itself. Ids come only from `save_evidence`, and that tool accepts a quote only if it appears word for word in a page fetched during the run. This is how the system stops **invented citations**.

4. Model fallback chain (reliability)



Any model that answers ends the chain. Failures: 429 / quota / 5xx / 404 / unsupported feature.

- `providers.FallbackModel` wraps the chain. A model that fails is put on **cooldown**: 10 min for quota, 60 s for rate limits and outages, 1 h for not-found or auth errors. Later calls skip it during the cooldown.
- Every switch is written to the run timeline as a `fallback` event, so the UI shows which model actually answered.
- The chain is set in `.env`: `RDI_MODEL_SMART=gemini-4-argon` puts argon first, and `RDI_FALLBACK_MODELS=gemini-3.6-flash,gemini-3.5-flash-lite` overrides the list. Set it to `none` to turn fallback off.

---

5. Agents

Agent	File	Model role	Output type	Tools
Research Planner	`agent_team/planner.py`	smart	`ResearchPlan`	—
Technical / Market / Risk specialists	`agent_team/specialists.py`	fast	`SpecialistFindings`	search_web, scrape_page, extract_claims, save_evidence, retrieve_evidence
Research Orchestrator	`agent_team/orchestrator.py`	smart	`OrchestratorReview`	delegate_technical / market / risk_research, compare_sources, retrieve_evidence
Report Writer	`agent_team/writer.py`	smart	`DecisionReport`	retrieve_evidence
Critic	`agent_team/critic.py`	smart	`Critique`	check_claim_support, retrieve_evidence
Research Assistant (chat)	`agent_team/assistant.py`	fast	`ChatReply`	retrieve_evidence, scrape_page, publish_report (protected)
Scope classifier	`agent_team/scope_classifier.py`	fast	`ScopeCheck`	—

All prompts are in `agent_team/prompts.py` and versioned. The prompt version is saved in the trace metadata.

---

6. The five agentic patterns

Pattern	Where	How
Prompt chaining	`workflows/pipeline.py`	Planner → typed `ResearchPlan` → each later step
Routing	`workflows/routing.py`, `services/policy.py::decide_path`	Python lookup table; a *quick* path for one low-impact sub-question; adds `SQ-RISK` if missing
Parallelization	`workflows/parallel_research.py`	`asyncio.gather(..., return_exceptions=True)` + semaphore (`RDI_MAX_PARALLEL`)
Orchestrator-workers	`workflows/orchestration.py`	The LLM chooses which `delegate_*` worker to call; Python enforces the budget of 3
Evaluator-optimizer	`workflows/evaluator_optimizer.py`	Writer ⇄ validator + Critic; stops on approve, max rounds, or no progress; keeps the best draft. If it is not approved, findings with numbers missing from their cited quotes are removed in code before human review

Code-orchestrated vs LLM-orchestrated decisions

Decision	Who decides	Why
Sub-question → specialist, mandatory risk review	**Python**	A safety rule must not depend on a prompt
Quick vs full path, human approval needed	**Python** (`decide_path`)	Business rule; must be auditable
Stop condition of the critique loop	**Python**	Major validator issues always force a revision
Is publishing allowed?	**Python** (`publish_report`)	Authorisation: approval + valid citations
How to break the question into sub-questions	**LLM** (Planner)	Needs interpretation
Which gap to fill / which worker to call	**LLM** (Orchestrator)	Flexible delegation
Is the question in scope?	**LLM** (scope classifier)	Needs language understanding

7. Safety and guardrails

Layer	What it catches	File
Input guardrail (deterministic)	"ignore previous instructions", system-prompt leaks, API keys and secrets, oversized input	`guardrails/input_guardrails.py`
Input guardrail (LLM)	Off-topic requests ("write a poem")	`scope_classifier.py`
Page sanitisation	Instructions hidden in web pages are redacted, the page is flagged, and its quality score is capped at 0.10	`services/injection.py`

Layer	What it catches	File
URL policy	localhost, `169.254.169.254`, `file://`, very long URLs	`services/policy.py::check_url_allowed`
Verbatim evidence	Quotes that do not appear in the fetched page	`services/evidence.py`
Output guardrail	Citations to unknown ids (e.g. `E99`), injected text in the report	`guardrails/output_guardrails.py`
Deterministic validator	Numbers that appear in no cited quote, claims without citations	`workflows/evaluator_optimizer.py`
Protected action	`publish_report` refuses to publish without approval when approval is required	`services/publisher.py`
Chat scope	Follow-up chat answers only questions about the loaded run; off-topic messages get a fixed refusal, and answers citing unknown evidence ids are replaced	`workflows/chat.py`

Human-in-the-loop rules (`workflows/hitl.py`)

A report goes to the approval queue, not straight to publishing, when **any** of these is true:

- the Planner rated the impact **high**;
- the question contains a sensitive keyword (legal, people or student records, investment, security …);
- major issues are still open after the last critic round (status `needs_human_review`);
- the Writer's confidence is **low**.

Otherwise the report is published automatically. Approvals are stored in SQLite and decided atomically, so the same approval cannot be decided twice.

8. Reliability

Failure	Handling
Invalid structured output	`runtime.safe_run` sends the validation error back to the model and retries
429 / quota	wait 20 s × attempt, plus switching to the next model in the fallback chain
Dead or flaky page	`fetch_with_retry` with backoff; a dead link is logged as `tool_failure` and the run continues
Search outage (demo checkbox)	the tool returns an error string (`failure_error_function`) and the agent carries on
One specialist crashes	`return_exceptions=True` lets the others finish; the run fails only if **all** specialists fail
Infinite loops	`max_turns=14` per agent, delegation budget 3, critic rounds 3 or 2

9. Project layout

```
app.py                        Gradio UI: Research · Approvals · Chat tabs
data/corpus.json              fictional web corpus (Delta University LMS decision) with traps
src/rdi/
  config.py                   providers, models, fallback chains, limits, sensitive keywords
  providers.py                provider clients, FallbackModel, local tracing
  runtime.py                  safe_run (retries, 429 waits), hooks, step timing
  models.py · context.py      Pydantic models · per-run context (evidence store, timeline)
  tools.py                    @function_tool definitions
  agent_team/                 prompts + one file per agent
  guardrails/                 input and output guardrails
  workflows/                  pipeline · routing · parallel_research · orchestration ·
                              evaluator_optimizer · hitl · chat
  services/                   deterministic layer: bm25, evidence, injection, policy,
                              approvals, publisher, search_backend, source_quality
  ui_theme.py · ui_helpers.py  UI rendering (pure functions, unit-tested)
scripts/                      check_provider.py · eval_agents.py · show_agents.py
tests/                        145 unit tests + 15 offline agent scenarios (fake SDK)
```

10. Setup and running

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
copy .env.example .env        # then edit .env (see below)
python scripts\check_provider.py
python -m pytest -q
python app.py                 # http://127.0.0.1:7860
```

Minimal `.env` (free Gemini tier):

```
RDI_PROVIDER=gemini
GEMINI_API_KEY=<your key from aistudio.google.com/apikey>
RDI_MODEL_SMART=gemini-4-argon
```

Never put a key in `.env.example` and never commit `.env`; it is listed in `.gitignore`.

11. Demo scenarios (brief §9)

#	Scenario	Trigger	Evidence on screen
1	Happy path	Example 2 (Atlas Open pilot)	✅ Published, Report tab
2	≥3 sequential steps	any full run	Timeline: search → scrape → extract → save
3	Routing	Example 1 (full) vs Example 3 (quick)	Plan & routing tab, `SQ-RISK` added by code
4	Parallelization	any full run	interleaved `parallel start/done` events

#	Scenario	Trigger	Evidence on screen
5	Orchestrator-workers	Example 1	delegations in the Research tab
6	Evaluator rejects	Fault = `unsupported_claim`	Evaluator tab: round 1 revise → round 2 approve
7	Input guardrail	Example 4, or "write a poem"	⛔ Blocked
8	Blocked action	Chat: `read http://169.254.169.254/latest`, or "publish" on a pending run	tool returns *blocked*
9	Output guardrail	Fault = `invented_citation`	round 1 rejected (E99)
10	Tool failure + recovery	tick *search outage*; dead/flaky pages	`tool_failure`, `tool_retry`, run completes
11	Human escalation	Example 1 (high impact)	⏸ Pending → approve in Approvals tab → published
12	Tracing	any run	Timeline & trace tab, `var/traces/<run_id>.jsonl`